BANGALORE INSTITUTE OF TECHNOLOGY

Department of Information Science & Engineering

COURSE SYLLABUS: COMPUTER NETWORKS

Course Code: BIS586 | Academic Year: 2026-2027

Overview: This comprehensive course covers computer networking from physical transmission media to application protocols, focusing on mathematical modeling and protocol analysis.

Unit 1: Introduction to Networks

· Network Types & Topologies: LAN, WAN, MAN, Star, Mesh, Ring, Bus topologies.

· OSI Reference Model: 7-layer architecture, encapsulation, protocol data units.

· TCP/IP Protocol Architecture: Four-layer internet model, RFC standards.

· Packet Switching vs Circuit Switching: Statistical multiplexing, queuing trade-offs.

Unit 2: Physical Layer

· Guided Transmission Media: Twisted pair, coaxial cable, single and multimode fiber.

· Digital Modulation & Encoding: Manchester, NRZ, QAM line coding schemes.

· Multiplexing Techniques: FDM, TDM, WDM wavelength division in optical fibers.

· Wireless Transmission & Antennas: Terrestrial microwave, line of sight, ISM bands.

Unit 3: Data Link Layer

· Framing & Flow Control: Byte stuffing, bit stuffing, sliding window, Go-Back-N.

· Error Detection & Correction: CRC polynomial arithmetic, Hamming distance bounds.

· Medium Access Control Protocols: Pure ALOHA, Slotted ALOHA, CSMA/CD, CSMA/CA.

· Ethernet & IEEE 802.3: 48-bit MAC addressing, preamble, collision domains.

Unit 4: Network Layer

· IPv4 Addressing & Subnetting: Classless Inter-Domain Routing (CIDR), VLSM, RFC 1918.

· Routing Algorithms: Dijkstra's Link-State (OSPF), Bellman-Ford Distance Vector (RIP).

· IPv6 Architecture & Transition: 128-bit address space, simplified base header.

· Congestion Control Principles: Leaky bucket, token bucket traffic shaping.

Unit 5: Transport Layer

· TCP 3-Way Handshake & Reliability: SYN, ACK, FIN flags, sequence numbers, 2MSL TIME_WAIT.

· UDP & Real-Time Applications: Connectionless 8-byte header overhead for multimedia.

· TCP Flow Control & Sliding Window: Receiver window (rwnd), buffer management.

· TCP Congestion Control (Reno, Cubic): Slow start, AIMD, fast retransmit, duplicate ACKs.

Unit 6: Application Layer

· Domain Name System (DNS): Root servers, TLDs, authoritative resolvers, port 53.

· HTTP/HTTPS & Web Protocols: HTTP/1.1 pipelining, HTTP/2 multiplexing, HTTP/3 QUIC.

· Email Protocols: SMTP message transfer, IMAP/POP3 mailbox retrieval.

· Network Security & TLS/SSL: Ephemeral Diffie-Hellman key exchange, certificates.